Janghyun **Noh**

AI Researcher

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Research Interest

Machine Learning, Deep Learning, Multimodal Learning, Medical AI, Bioinformatics

Education

Myongji University — *Seoul, S.Korea*

M.S. in Artificial Intelligence — *Sep. 2024 - Aug. 2026*

- Conducted research at the Artificial Intelligence and Data Analytics Laboratory under the supervision of Prof. Minsik Oh.
- Conducted collaborative research with external institutions including Seoul National University Hospital (SNUH), the Korea Disease Control and Prevention Agency (KDCA), Kyungpook National University, and AIgenDrug Co., Ltd.
- Served as a teaching assistant for undergraduate courses including Deep Learning, Machine Learning, Data Mining, and Programming (Fall 2024 – Spring 2026).

Myongji University — *Seoul, S.Korea*

B.S. in Convergence Software and Data Technology — *Mar. 2018 - Aug. 2024*

- Conducted undergraduate research at the Artificial Intelligence and Data Analytics Laboratory under the supervision of Prof. Minsik Oh.
- Conducted collaborative research on clinical and genetic factors associated with hypertension progression in patients with hyperlipidemia in collaboration with the Korea Disease Control and Prevention Agency (KDCA).

RED School – Alternative Middle-High School — *Geumsan, S.Korea*

Student — *Mar. 2012 - Jan. 2018*

- Received academic excellence scholarships for two consecutive years.
- Participated in student leadership and community activities.

Research Experience

Artificial Intelligence and Data Analytics Lab, Myongji University — *Seoul, S.Korea*

Undergraduate Research Intern / Graduate Researcher — *Dec. 2022 - Aug. 2026*

- Developed scCap, an annotation-free phenotype prediction framework for single-cell RNA-seq data (first-author paper in Briefings in Bioinformatics), released with code and a Docker image.
- Built scRNA-seq models for MSI/MSS phenotype prediction and applied them to AI-based target discovery (with AIgenDrug).
- Conducted spatial transcriptomics analyses of TNBC Xenium datasets for cellular niche and tumor microenvironment characterization (with SNUH).
- Compared cirrhotic and non-cirrhotic HCC tissues using GeoMx spatial transcriptomics for immunomodulatory target discovery.
- Built a Biomni-based AI agent environment for analyzing CRISPR-Cas9-induced large DNA deletions.
- Developed multimodal COVID-19 severity prediction models combining MIMIC-IV clinical data and MIMIC-CXR chest X-rays (with KDCA).
- Implemented an nf-core/sarek variant-calling pipeline for apple whole-genome sequencing data (with Kyungpook National University).
- Designed measurement-based labels and prediction models for hypertension progression in KoGES cohort data (with KDCA).

Publications & Presentations

Journal Articles

Annotation-free phenotype prediction using knowledge-augmented clustering from single-cell RNA sequencing data — *Briefings in Bioinformatics (BIB)*

Published — *2026*

J. Noh, Y. Shin, M. Kim, and M. Oh

Ferroptosis-based therapeutic strategy targeting iron metabolism and SLC7A11 overcomes oxaliplatin resistance in patient-derived pancreatic cancer organoids — *Journal of Experimental & Clinical Cancer Research (JECCR)*

Published — *2026*

J. Choi, K. Lee, M. Oh, **J. Noh**, Y. Kim, E. Hwang, I. Cho, W. Paik, J. Ryu, J. Ku, and S. Lee

Conference Presentations

Patient-Level Metacell Construction with Pre-trained Embeddings for Phenotype Prediction from scRNA-seq — *KIISE Korea Computer Congress (KCC)* — *2026*

Oral Presentation

Y. Shin, **J. Noh**, and M. Oh

Lightweight Analysis of Patient Phenotype Prediction Based on Single-Cell RNA-seq Data Reduction Techniques — *KIISE Korea Software Congress (KSC)* — *2025*

Poster Presentation (Best Poster Award)

Y. Shin, **J. Noh**, and M. Oh

An annotation-free framework for phenotype prediction using hierarchical multiple instance learning with single-cell RNA-sequencing data — *International Conference on Genome Informatics ISCB-Asia* — *2025*

Poster Presentation

J. Noh, Y. Shin, M. Kim, and M. Oh

Patient Phenotype Prediction from Single-cell RNA-seq Data through Annotation-free Multiple Instance Learning — *BIOINFO 2025, Korean Society for Bioinformatics (KSBI)* — *2025*

Poster Presentation (Excellent Poster Award)

J. Noh, Y. Shin, M. Kim, and M. Oh

AMIL: Automated Cell Grouping Framework for Phenotype Prediction with Multiple Instance Learning — *KIISE Korea Computer Congress (KCC)* — *2025*

Oral Presentation (Best Presentation Award)

J. Noh, M. Kim, and M. Oh

Predicting In-Hospital Mortality in Patients with Acute Respiratory Distress Syndrome Using MIMIC-III Data: A Comparative Study of Machine Learning Models with Varied Missing Data Handling Strategies — *KIISE Korea Software Congress (KSC)* — *2023*

Poster Presentation

J. Noh, G. Choi, and M. Oh

Honors & Awards

Awards

2026	**Research Honor Award**, Myongji University	*Seoul, S.Korea*
2025	**Best Poster Award**, KIISE Korea Software Congress (KSC)	*Seoul, S.Korea*
2025	**Excellent Poster Award**, BIOINFO 2025, Korean Society for Bioinformatics (KSBI)	*Seoul, S.Korea*
2025	**Best Presentation Award**, KIISE Korea Computer Congress (KCC)	*Seoul, S.Korea*
2022	**President's Merit Award**, Korea Trade-Investment Promotion Agency (KOTRA)	*Seoul, S.Korea*

Scholarships

2022	**TriBIG Project Scholarship**, Korea Trade-Investment Promotion Agency (KOTRA)	*Seoul, S.Korea*
2019	**Globalization Scholarship**, Myongji University	*Seoul, S.Korea*
2018	**Scholarship III**, Myongji University	*Seoul, S.Korea*
2016	**Academic Excellence Scholarship**, RED School	*Geumsan, S.Korea*
2015	**Academic Excellence Scholarship**, RED School	*Geumsan, S.Korea*

Projects

2024 – 2026	AI-based target discovery platform for CAR-Treg cell therapy in chronic inflammatory disease
2024 – 2026	Genetic and physiological mechanisms of apple blossom dormancy release under climate change
2024 – 2026	Spatial transcriptomics-based target discovery and validation for refractory solid tumors
2025 – 2026	Client-side web platform for detecting large CRISPR-Cas9-induced DNA deletions
2025 – 2026	GeoMx spatial transcriptomics of cirrhotic vs. non-cirrhotic HCC tissue
2025	COVID-19 severity prediction using multimodal clinical and imaging data
2023 – 2024	Hypertension-progression risk groups in hyperlipidemia patients (KoGES cohort)
2023	CatchYou: criminal composite sketch generation from witness descriptions
2022	Customs notice analysis and monitoring system for global supply-chain crises

Technical Skills

Programming	Python, R
Frameworks	PyTorch, TensorFlow, Scikit-learn, XGBoost, Scanpy, FastAPI
Tools	Linux, Git, Docker, Nextflow
Research	Machine Learning, Deep Learning, Multimodal Learning, Medical AI, Bioinformatics
Languages	Korean (Native), English (Full Professional Proficiency; OPIc IH 2026, TOEIC 920 2023)
Certifications	Engineer Information Processing (2023), SQL Developer – SQLD (2023)

Extracurricular Activities

Domestic

Google Developer Student Clubs, Myongji University — *Seoul, S.Korea*

AI Team Member — *Oct. 2022 - Dec. 2023*

- Engaged in machine learning and deep learning study groups
- Collaborated on AI projects and participated in AI competitions

Big Leader AI Academy — *Seoul, S.Korea*

Project Member — *Jul. 2022 - Oct. 2022*

- Developed a customs notice analysis and monitoring system in collaboration with the Big Data Team of KOTRA
- Participated in an industry-collaborative AI project for KOTRA internal service applications (TriBIG)
- Received a scholarship through outstanding performance in the TriBIG project

Data Youth Campus, Dongguk University — *Seoul, S.Korea*

Participant — *Jul. 2022 - Aug. 2022*

- Completed a data science-based intelligent software program affiliated with Korea Data Agency (KDATA)
- Conducted practical projects and training in machine learning and data analysis

International

Camino de Santiago (French Way) — *Spain*

Pilgrim — *Sep. 2026 - Oct. 2026*

Medical Mission Trip to the Philippines (GMMA) — *Manila, Philippines*

Volunteer — *Jan. 2024*

- Participated in medical volunteer activities with international healthcare professionals

International Work Camp in Iceland — *Reykjavik, Iceland*

Volunteer — *Dec. 2019 - Jan. 2020*

- Participated in environmental campaigns and intercultural exchange with international volunteers

Mission Trip to Vietnam — *Hanoi, Vietnam*

Volunteer — *Aug. 2018*

- Participated in intercultural exchange and community volunteer activities